

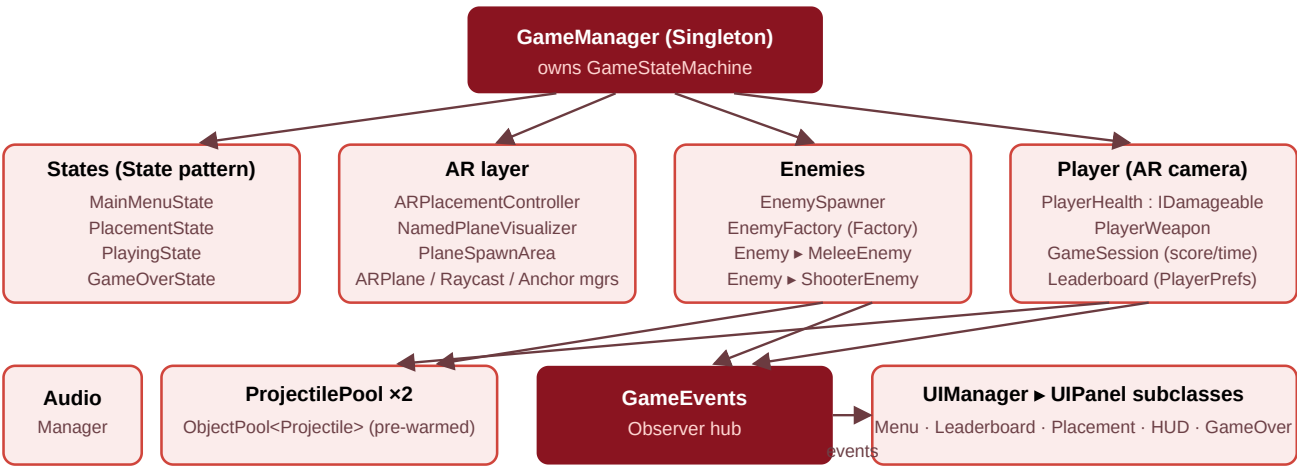
Holdfast AR: Technical Documentation

Mobile AR survival shooter · Portrait · Unity 6 + AR Foundation (ARCore / ARKit) · Developer: **Joshua Chukwuebuka Moses**

Story: you are the last colonist on the frontier world Kepler-9. Your landing dome has come down on the only flat ground you can find (the real floor), and the planet's natives, alien grunts and armoured bug mechs, are closing in. Hold the dome with your ray gun until the evac ship arrives (the match timer).

1. Architecture overview

The game is split into small systems that talk through a **GameManager** (the coordinator) and a static **GameEvents** hub. Gameplay code never references UI classes: it raises events, and the HUD listens. The match flow is driven by a **state machine**: `MainMenu` → `Placement` → `Playing` → `GameOver`, and Restart jumps straight back to `Playing`.



2. OOP structure

Principle	Where it is applied
Encapsulation	State lives in private fields with read-only properties: <code>GameSession.Score</code> only changes through <code>RegisterKill()</code> , <code>PlayerHealth.Current</code> only through <code>TakeDamage()</code> / <code>ResetHealth()</code> , and <code>Leaderboard</code> hides its JSON/PlayerPrefs storage behind <code>Add()</code> and <code>Sessions.Inspector</code> values use <code>[SerializeField] private</code> .
Abstraction	<code>IDamageable</code> (Team, IsAlive, TakeDamage) lets projectiles and melee attacks hurt anything without knowing concrete types. <code>IPoolable</code> abstracts pool reset hooks. Abstract classes <code>Enemy</code> , <code>GameState</code> and <code>UIPanel</code> define the contract without the details.
Inheritance	<code>Enemy</code> → <code>MeleeEnemy</code> , <code>ShooterEnemy</code> share health, chasing, facing, cooldown, hit feedback and death. <code>GameState</code> → 4 states; <code>UIPanel</code> → 5 panels; <code>Singleton<T></code> → <code>GameManager</code> , <code>AudioManager</code> .
Polymorphism	<code>Enemy.Update()</code> calls the abstract <code>StoppingDistance</code> and <code>Attack()</code> : the Brute stops at 0.35 m and swings, while the Sentinel stops at 1.4 m and fires. Both drive the same Animator parameters (<code>Speed</code> , <code>Attack</code> , <code>Hit</code> , <code>Die</code>), so each model plays its own walk, punch or shoot, hit and death clips through one shared code path. The state machine only calls <code>Enter/Tick/Exit</code> , and <code>UIManager</code> only calls <code>Build/Show/Hide/Tick</code> on any panel.

Enemy comparison

	Melee "Brute"	Shooter "Sentinel"
Model	Green alien grunt (animated)	Yellow bug mech (animated)
Bullets to kill	3	5
Stops at	0.35 m	1.4 m
Attack range	0.5 m (contact)	2.2 m (projectile)
Damage / cooldown	15 / 1.1 s	8 per shot / 1.6 s
Score	100	150

Difficulty (bonus)

	Easy	Normal	Hard
Match length	60 s	90 s	120 s
Player HP	150	100	80
Spawn interval	3.2 s	2.4 s	1.6 s
Max enemies	4	6	9
Speed / damage ×	0.8 / 0.6	1 / 1	1.3 / 1.4
Score ×	0.8	1.0	1.5

Each level is a `DifficultySettings` `ScriptableObject` asset, so balancing needs no code changes.

3. Design patterns used and why

Pattern	Classes	Why
Object Pool	<code>ObjectPool<T></code> , <code>ProjectilePool</code> , <code>Projectile</code> : <code>IPoolable</code>	Bullets are fired many times per second. Creating/destroying them causes GC spikes and frame drops on phones, which is especially bad for AR tracking.
State	<code>GameState</code> , <code>GameStateMachine</code> , 4 concrete states	Each phase owns its setup/teardown (e.g. <code>PlayingState.Exit()</code> wipes enemies and bullets), which avoids a large switch and makes <code>Start</code> → <code>Play</code> → <code>End</code> transitions explicit.
Singleton	<code>Singleton<T></code> → <code>GameManager</code> , <code>AudioManager</code>	Exactly one coordinator and one audio owner, reachable from anywhere; duplicates self-destroy.
Factory	<code>EnemyFactory</code>	Only one class maps <code>EnemyType</code> → prefab and injects dependencies (<code>EnemyContext</code> : target, difficulty, enemy bullet pool). New enemy types need no spawner changes.
Observer	<code>GameEvents</code> , <code>HudPanel</code> , <code>SpawnMarkers</code> , <code>PlayingState</code>	Health, score, kills, time and enemy spawns are broadcast; <code>SpawnMarkers</code> listens to <code>EnemySpawned</code> to show a pin over each new enemy, or an edge arrow when it is off screen. The HUD subscribes in <code>OnShow</code> and unsubscribes in <code>OnHide</code> , so gameplay stays independent of UI.

4. Object Pool implementation

- **Pre-initialised:** `ProjectilePool.Awake()` creates `ObjectPool<Projectile>`, which instantiates all objects up front (40 player bullets, 30 enemy bullets), deactivates them and pushes them onto a `Stack`. This is the only `Instantiate` for bullets.
- **Get:** pops an inactive bullet, sets position/rotation, activates it and calls `OnTakenFromPool()`. If the pool is empty, the **oldest active bullet is recycled** (tracked in a `LinkedList`) instead of allocating, so there is never an `Instantiate` during gameplay.
- **Release:** on hit or after its 3 s lifetime, the bullet calls back to its owner pool: `OnReturnedToPool()` → deactivate → push. Double releases are ignored.
- **Proper reset:** age, speed, damage, direction and team are cleared, and the `TrailRenderer` is cleared and toggled so no streak is drawn from the previous position. `ReleaseAll()` runs when a match ends.
- **Hit detection:** each frame the bullet sphere-casts along its path (no tunnelling at AR scale) and damages the nearest `IDamageable` whose `Team` differs from the shooter's.

5. Sound system structure and sources

`AudioManager` (Singleton) is the **only owner of AudioSources**. Enemies and bullets have none, which avoids duplicate components: one 2D source for UI/player sounds (`PlayOneShot`, so rapid shots overlap on one source), one looping 2D music source, and a fixed round-robin pool of **6 spatial 3D sources** (linear roll-off 0.3–6 m, tuned for room-scale AR) that are moved to the enemy's position for positional sounds. Clips load from `Resources/Audio` by the `SoundId` enum name, with per-sound volume and a slight random pitch to avoid repetition.

Event (required ★)	SoundId	Triggered in	2D / 3D
★ Player shoot	PlayerShoot	<code>PlayerWeapon.Fire()</code>	2D
★ Player death	PlayerDeath	<code>PlayerHealth.TakeDamage()</code> at 0 HP	2D
★ Enemy spawn	EnemySpawn	<code>EnemySpawner.SpawnOne()</code>	3D
★ Enemy shoot (Shooter)	EnemyShoot	<code>ShooterEnemy.Attack()</code>	3D
★ Enemy damage (Melee attack)	MeleeAttack	<code>MeleeEnemy.Attack()</code>	3D
Player hurt · Enemy hit · Enemy death	PlayerHurt · EnemyHit · EnemyDeath	PlayerHealth · Enemy	2D · 3D · 3D
UI click · Arena placed · Victory · Ambient	UIClick · Place · Victory · Ambient	UIFactory · ARPlacementController · GameOverState · MainMenuState	2D · 3D · 2D · loop

Sources: all 12 clips are original and were synthesised by `Tools/generate_audio.py` (sine/square/saw oscillators, pitch sweeps, filtered noise, envelopes; 16-bit mono 44.1 kHz WAV). The ambient loop uses only frequencies that complete whole cycles in 16 s, so it loops seamlessly.

6. AR implementation notes

- **Plane detection:** `ARPlaneManager.requestedDetectionMode = Horizontal`. The default visualiser is replaced by `NamedPlaneVisualizer`, which triangulates each plane's boundary into its own mesh and maps UVs in metres (0.6 m per tile), so the texture reading **JOSHUA CHUKWUEBUKA MOSES** tiles across the real floor. It only renders while the plane is `Tracking` and not subsumed.
- **Tap-to-place:** `ARRaycastManager.Raycast(PlaneWithinPolygon) → ARAnchorManager.AttachAnchor(plane, pose)`, and the arena is parented to the anchor. After placement, later taps are ignored, the plane manager is disabled (no new planes) and trackers are hidden.
- **Spawning on planes:** at placement, `PlaneSpawnArea` stores every horizontal plane polygon in arena-local space. Spawns pick random points 0.9–2.2 m from the base and at least 1.1 m from the player, then keep only points inside a real plane polygon.
- **Portrait UI:** every panel is built in code inside a 1080×1920 portrait frame fitted to the device safe area, using the Kenney UI Pack (Space Expansion) sprites (9-sliced, tinted to the red space palette) and the Kenney Future font. The FIRE button sits bottom-right, clear of the centred ray gun, and the gun is hidden outside a match.
- **Leaderboard:** `SessionRecord` list serialised with `JsonUtility` into `PlayerPrefs`; the newest record is inserted first and the list is trimmed to 5.

7. Art assets and sources

Asset	Used for	Source · licence
Ultimate Space Kit by Quaternius	Brute (Enemy Large), Sentinel (Mech), dome, rocks, plants, solar panel	poly.pizza · CC0
Sci-Fi Gun Pack by Quaternius	Player ray gun	poly.pizza · CC0
UI Pack (Space Expansion) by Kenney	Panels, buttons, bars, crosshair, spawn markers	kenney.nl · CC0
Kenney Future font	All UI text and the plane tracker label	kenney.nl · CC0

The app icon is a render of the Brute model framed by a red reticle. Animator Controllers, materials, the plane tracker texture and all sounds were made for this project.